

Matrices and Their Operations

ECON 320 – Introduction to Mathematical Economics

Fall 2025

1 Introduction

A **matrix** is a rectangular array of numbers arranged in rows and columns. They provide a compact representation of systems of linear equations, econometric models, and linear transformations. Matrices are widely used in economics: equilibrium analysis, input–output models, comparative statics, and econometric estimation.

2 Types of Matrices

- **Row Matrix:** One row only. Example: $[3 \ 5 \ 7]$.
- **Column Matrix:** One column only. Example: $\begin{bmatrix} 2 \\ 4 \\ 6 \end{bmatrix}$.
- **Square Matrix:** Same number of rows and columns ($n \times n$).
- **Diagonal Matrix:** All off-diagonal entries are zero.
- **Identity Matrix I :** Diagonal matrix with 1's on the diagonal. Acts like 1 in multiplication.
- **Zero Matrix 0 :** All entries are zero.
- **Symmetric Matrix:** $A = A'$.

3 Basic Operations

3.1 Addition and Subtraction

Defined element-wise, only possible if matrices have the same dimensions.

3.2 Scalar Multiplication

Multiply every entry of a matrix by the scalar.

3.3 Matrix Multiplication

If A is $m \times n$ and B is $n \times p$, then AB is $m \times p$:

$$(AB)_{ij} = \sum_{k=1}^n a_{ik}b_{kj}.$$

3.4 Transpose

The transpose A' (or A^T) is obtained by flipping rows into columns. Properties:

$$(A')' = A, \quad (A + B)' = A' + B', \quad (AB)' = B' A'.$$

4 Determinants

The determinant is a scalar value associated with square matrices.

4.1 Definition

For a 2×2 matrix:

$$\det \begin{bmatrix} a & b \\ c & d \end{bmatrix} = ad - bc.$$

For higher-order matrices, expansion by minors or Laplace expansion is used.

4.2 Properties

- $\det(AB) = \det(A) \det(B)$.
- $\det(A') = \det(A)$.
- If $\det(A) = 0$, A is **singular** and not invertible.

5 Inverse of a Matrix

The inverse A^{-1} (if it exists) satisfies:

$$AA^{-1} = A^{-1}A = I.$$

For a 2×2 matrix:

$$A^{-1} = \frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}.$$

6 Jacobian Matrix

The **Jacobian** of a vector-valued function $F : \mathbb{R}^n \rightarrow \mathbb{R}^m$ is the $m \times n$ matrix of first-order partial derivatives:

$$J(x) = \begin{bmatrix} \frac{\partial f_1}{\partial x_1} & \cdots & \frac{\partial f_1}{\partial x_n} \\ \vdots & \ddots & \vdots \\ \frac{\partial f_m}{\partial x_1} & \cdots & \frac{\partial f_m}{\partial x_n} \end{bmatrix}.$$

In economics, Jacobians are used in comparative statics, general equilibrium, and optimization problems.

Examples

6.1 Determinant Example

Let

$$A = \begin{bmatrix} 2 & 1 \\ 3 & 4 \end{bmatrix}.$$

Compute $\det(A)$:

$$\det(A) = (2)(4) - (1)(3) = 8 - 3 = 5.$$

Since $\det(A) \neq 0$, A is invertible.

6.2 Jacobian Example

Consider the demand system:

$$q_1(p_1, p_2) = 10 - 2p_1 + p_2, \quad q_2(p_1, p_2) = 20 + p_1 - 3p_2.$$

The Jacobian is:

$$J = \begin{bmatrix} \frac{\partial q_1}{\partial p_1} & \frac{\partial q_1}{\partial p_2} \\ \frac{\partial q_2}{\partial p_1} & \frac{\partial q_2}{\partial p_2} \end{bmatrix} = \begin{bmatrix} -2 & 1 \\ 1 & -3 \end{bmatrix}.$$